

Problem Setup

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Problem Setup

We are given the quadratic equation

$$x^2 + 2x + 1 = 0$$

and we are asked to find all real (or complex) values of x that satisfy it.

Strategy and Key Principles

The equation is a quadratic polynomial. We can solve it by either:

- Factoring the perfect-square trinomial, or
- Completing the square (which is equivalent to factoring in this case).

Both methods rely on the identity $(a + b)^2 = a^2 + 2ab + b^2$.

Detailed Solution

First recognise the left-hand side as a perfect square:

$$x^2 + 2x + 1 = (x + 1)^2.$$

Thus the equation becomes

$$(x + 1)^2 = 0.$$

Taking the square root of both sides (noting that the square root of zero is zero) gives

$$x + 1 = 0.$$

Finally, isolate x :

$$x = -1.$$

For completeness, we can display the algebraic manipulation in an aligned environment:

$$\begin{aligned} x^2 + 2x + 1 &= 0 \\ (x + 1)^2 &= 0 \\ x + 1 &= 0 \\ x &= -1 \end{aligned}$$

Final Answer

$$\boxed{x = -1}$$

Key Insights

- The quadratic $x^2 + 2x + 1$ is a perfect square; recognizing this avoids unnecessary use of the quadratic formula.
- When a squared expression equals zero, the expression itself must be zero—there is no “plus/minus” ambiguity.
- A common mistake is to forget that $(x + 1)^2 = 0$ yields a single root $x = -1$, not two distinct roots.